

Experiment 1

WIRELESS TECHNOLOGY

1. Bluetooth

Overview: Bluetooth is a short-range wireless technology used to exchange data between devices like smartphones, laptops, and wearables. It operates in the 2.4 GHz ISM band and is widely used for personal area networks (PANs). Bluetooth is ideal for low-power, short-range communication, and its low energy variant, Bluetooth Low Energy (BLE), is specifically designed for battery-powered devices.

Common Uses: Connecting wireless headphones, speakers, keyboards, mice, fitness trackers, smartwatches, and more.

Range: 10 meters (Standard), up to 100 meters (Bluetooth 5.0)

- **Cost:** Low
 - **Propagation Delay:** Low (typically 1–2 ms)
 - **Power Consumption:** Low (especially in Bluetooth Low Energy)
 - **Throughput:** Up to 3 Mbps (Classic Bluetooth), Up to 2 Mbps (Bluetooth Low Energy)
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2. Wi-Fi

Overview: Wi-Fi is a wireless communication technology used primarily to provide internet access within a local area network (LAN). It operates on the 2.4 GHz, 5 GHz, and 6 GHz frequency bands (Wi-Fi 6/6E). Wi-Fi is a crucial technology for high-speed internet connectivity in homes, businesses, and public spaces.

Common Uses: Internet access in homes, offices, coffee shops, and public spaces, as well as for connecting devices like laptops, smartphones, and IoT devices to the internet.

- **Range:** 30–100 meters (indoor), up to 300 meters (outdoor)
 - **Cost:** Moderate
 - **Propagation Delay:** Medium (around 10–50 ms)
 - **Power Consumption:** Moderate to High (depends on use and Wi-Fi standard)
 - **Throughput:** Up to 9.6 Gbps (Wi-Fi 6)
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3. Zigbee

Overview: Zigbee is a low-power, short-range wireless communication technology based on the IEEE 802.15.4 standard. It is primarily used for IoT applications, home automation, and smart energy solutions. Zigbee is designed for low data-rate, low-power, and low-cost devices, and supports mesh networking for greater range and reliability.

Common Uses: Home automation (lighting, thermostats, door locks), industrial automation, smart energy meters, and remote sensors.

- **Range:** 10–100 meters
 - **Cost:** Low
 - **Propagation Delay:** Low (typically 10–100 ms)
 - **Power Consumption:** Very Low
 - **Throughput:** 250 kbps (Zigbee 3.0)
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4. LoRa (Long Range)

Overview: LoRa is a long-range, low-power wireless communication protocol designed for Internet of Things (IoT) applications. It operates in the sub-gigahertz frequency bands (such as 433 MHz, 868 MHz, and 915 MHz) and is capable of communicating over long distances, especially in rural or wide-area deployments. LoRa is typically used in conjunction with LoRaWAN (LoRa Wide Area Network), which defines the communication protocol for LoRa networks.

Common Uses: IoT applications like agriculture monitoring, smart cities, environmental sensors, and asset tracking.

- **Range:** Up to 15 km in rural areas, 2–5 km in urban areas
 - **Cost:** Low to Moderate
 - **Propagation Delay:** Medium (varies based on range and environment)
 - **Power Consumption:** Very Low
 - **Throughput:** 50–500 kbps
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5. 5G (Fifth Generation)

Overview: 5G is the latest generation of cellular mobile networks, offering significantly higher speeds, lower latency, and greater reliability than previous generations (4G, 3G). It utilizes a mix of low, mid, and high-frequency bands (including mmWave), enabling faster data rates, enhanced mobile broadband, and

support for a massive number of connected devices. 5G also supports ultra-reliable low-latency communication (URLLC) and massive machine-type communication (mMTC).

Common Uses: High-speed mobile internet, autonomous vehicles, industrial automation, smart cities, telemedicine, augmented and virtual reality, and more.

- **Range:** 100 meters (low band), up to 1 km (mid band), and less than 100 meters (high band or mmWave)
 - **Cost:** High (due to infrastructure costs)
 - **Propagation Delay:** Very Low (around 1–10 ms)
 - **Power Consumption:** Moderate (depends on network usage)
 - **Throughput:** 10 Gbps (theoretical maximum)
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6. NFC (Near Field Communication)

Overview: NFC is a short-range wireless communication technology designed for very close proximity interactions (a few centimeters or less). It operates at 13.56 MHz and enables devices to exchange data when they are near each other, often through passive (unpowered) tags. NFC is widely used for secure transactions, access control, and contactless payment systems.

Common Uses: Contactless payments (e.g., Apple Pay, Google Pay), public transportation fare systems, digital tickets, access control, and data exchange between devices (e.g., pairing devices).

- **Range:** 0–10 cm
 - **Cost:** Low
 - **Propagation Delay:** Very Low (milliseconds)
 - **Power Consumption:** Very Low
 - **Throughput:** 424 kbps
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7. Infrared (IR)

Overview: Infrared is a short-range wireless communication technology that uses infrared light waves to transmit data. It typically operates in the infrared spectrum (around 850 nm to 950 nm wavelength). IR is mainly used in consumer electronics for device control (e.g., remote controls for TVs, air conditioners).

Common Uses: Remote controls, device pairing, short-range communication between devices, and data transfer over short distances.

- **Range:** Typically 1–5 meters
- **Cost:** Low

- **Propagation Delay:** Very Low (milliseconds)
 - **Power Consumption:** Very Low
 - **Throughput:** Up to 4 Mbps
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8. Satellite Communication

Overview: Satellite communication uses communication satellites to transmit and receive signals, enabling long-range, global communication. Signals are transmitted from ground stations to satellites in geostationary or low-Earth orbits, which relay the signals back to other stations on Earth. Satellite communication is widely used for broadcasting, global telecommunication, and remote areas where terrestrial infrastructure is unavailable.

Common Uses: GPS, global telephony, satellite TV, internet access in remote areas, weather forecasting, and military communication.

- **Range:** Global (depends on satellite orbit)
 - **Cost:** High (due to infrastructure and satellite launch)
 - **Propagation Delay:** High (250 ms–1 second)
 - **Power Consumption:** High (for transmission)
 - **Throughput:** 1–50 Mbps (depending on satellite and equipment)
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9. Cellular Communication

Overview: Cellular communication refers to the use of mobile networks to provide voice, text, and data services. Different generations of cellular networks (3G, 4G, and 5G) offer increasing speeds, capacity, and capabilities.

- **3G** (third generation) provided improvements in data speeds, enabling mobile internet and multimedia services.
- **4G** (fourth generation) offers significantly faster internet speeds, supporting HD video streaming, gaming, and other high-demand services.
- **5G** (fifth generation) delivers ultra-fast internet speeds, low latency, and supports massive IoT device connectivity.

Common Uses: Mobile phone services, mobile internet, video streaming, remote healthcare, connected vehicles, and IoT.

- **Range:**
 - 3G: Up to several kilometers (cell radius)
 - 4G: 5–10 km (urban)
 - 5G: 100 meters (mmWave), 1 km (low/mid band)

- **Cost:** Moderate to High (varies by generation)
- **Propagation Delay:**
 - 3G: 50–100 ms
 - 4G: 20–40 ms
 - 5G: <10 ms
- **Power Consumption:**
 - 3G: Moderate
 - 4G: Moderate
 - 5G: High (especially in mmWave mode)
- **Throughput:**
 - 3G: 2 Mbps
 - 4G: 100 Mbps–1 Gbps
 - 5G: Up to 10 Gbps (theoretical)

Table of Key Parameters

Parameter	Bluetooth	Wi-Fi	Zigbee	LoRa	5G	NFC	Infrared	Satellite Communication	Cellular (3G/4/5G)
Range	10–100 meters	30–300 meters	10–100 meters	2–15 km	100m (low band), 1 km (mid band), <100m (mmWave)	0–10 cm	1–5 meters	Global	3G: Up to several km, 4G: 5–10 km, 5G: 100m–1 km
Cost	Low ₹(50 – 1,500)	Moderate ₹(500 –10,000)	Low ₹(200 – 2500)	Low-Moderate ₹(500-5000)	High ₹(2000 – 50000+)	Low ₹(100 – 2000 +)	Low ₹(10 – 1500)	High ₹(10000 – 200000+)	Moderate-high ₹(500-40000)
Propagation Delay	1–2 ms	10–50 ms	10–100 ms	Medium	1–10 ms	Very Low	Very Low	250 ms–1 second	3G: 50–100 ms, 4G: 20–40 ms, 5G: <10 ms

Power Consumption	Low	Moderate	Very Low	Very Low	Moderate	Very Low	Very Low	High	3G: Moderate, 4G: Moderate, 5G: High
Throughput	2–3 Mbps (Classic), 2 Mbps (BLE)	Up to 9.6 Gbps	250 kbps	50–500 kbps	10 Gbps (theoretical)	424 kbps	Up to 4 Mbps	1–50 Mbps	3G: 2 Mbps, 4G: 100 Mbps–1 Gbps, 5G: Up to 10 Gbps